

HW06 — API Testing Report

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Môn: Kiểm thử phần mềm — HW06 (AI-first API Testing)
SUT: EShop (<http://localhost:3000>, Node.js + Express + SQLite)
Công cụ: Postman · Newman 6.2.2 · newman-reporter-htmlextra · GitHub Actions
Ngày: 2026-08-19
Máy: Spartans-MB-Pro---C2F3HXFHW (Apple M3 Pro, macOS)
Repo: <https://github.com/yuran1811/hcmus-sw-testing--eshop-sut/tree/hw6/23127152>

1. Introduction

Bài tập yêu cầu pipeline kiểm thử API theo Bloom-AI: **generate** → **audit** → **extend** → **execute** → **report bugs**, kèm Postman/Newman, CI/CD, Agent Skill (G9.5) và AI Audit/Critique.

Ba API đã khóa (không trùng nhóm):

Pool	Feature	Endpoint(s)
A	FR-05 Product listing & search	GET /api/products
B	FR-11 Order history + detail	GET /api/orders/my-orders, GET /api/orders/:id
C	FR-15 Product management CRUD	POST/PUT/DELETE /api/products

Chi tiết Phase 0: [PHASE0_PLANNING.md](#).

2. API Selection

Pool	Feature	Endpoint(s)	Rationale
A	FR-05 Product listing & search	GET /api/products	Public API; domain partitions trên search; bề mặt SEC-05 SQLi
B	FR-11 Order history + detail	GET /api/orders/my-orders, GET /api/orders/:id	Cấp JWT vs no-auth; kiểm IDOR / ownership / PII
C	FR-15 Product management CRUD	POST/PUT/DELETE /api/products	Admin CRUD theo spec; SEC-02/SEC-03

3. API 1 — Pool A (FR-05)

3.1 AI Generation Process

Sinh TC **từng bước** (skill api-test-generate), không một prompt chung:

Step	Nội dung	TC IDs
A Domain	Valid/invalid/boundary/charset/encoding cho search	TC-A1-001...018
B State	N/A state machine + 2 case idempotent GET	TC-A1-019...020
C Security	SEC-01...07, trọng tâm SEC-05 SQLi	TC-A1-021...030
D Schema	Content-Type, array root, fields/types, error ≠ HTML	TC-A1-031...040

Tổng AI: 40 TC (≥ 35) → [test-cases/API1 PoolA/generated.md](#).

3.2 Audit Summary

Human audit (skill api-test-audit): **32 VALID / 3 INVALID / 5 INCOMPLETE** → [audit.md](#).

Ví dụ sửa: pin hành vì search= rỗng (= list-all); expected SEC-05 viết theo **spec an toàn**, không theo SUT đang lỗi.

3.3 Extended Test Cases

6 TC manual (skill api-test-extend) → [extended.md](#):

TC	Gap AI miss
TC-A1-E01...E04	URL-encoded / double-encoded SQLi; Content-Type trên lỗi
TC-A1-E03	Oracle so sánh số dòng baseline vs tautology
TC-A1-E05...E06	Sequence sau DROP probe; assert không có <h1>Database Error

Excel: [API1 PoolA.xlsx](#).

3.4 Execution Results

Newman folder **API1 — FR-05** trên <http://localhost:3000>:

Metric	Value
Requests	20
Assertions	26
Passed	22
Failed	4

- HTML: [postman/reports/newman-api1-fr05.html](#)
- CLI: [postman/screenshots/newman-api1-fr05-cli.txt](#)
- Evidence header: [postman-prerequest.png](#), [postman-console-x-student-id.png](#), [newman-cli.png](#)

4 fail = SEC-05 (SQLi dump / HTML 500) — đúng kỳ vọng secure behavior.

3.5 Bugs Found

Bug	Severity	GitHub Issue	Evidence
BUG-PRODUCTS-001 SQLi search dumps catalog	Critical	#300	bug-reports/screenshots/BUG-PRODUCTS-001-sqli.json, issue-300-*.png
BUG-PRODUCTS-002 HTML 500 on bad SQL	Major	#301	BUG-PRODUCTS-002-body.html, issue-301-*.png

Author issues: **Anhnguyenk835**.

4. API 2 — Pool B (FR-11)

4.1 AI Generation Process

Steps A–D cho my-orders (JWT) + GET /api/orders/:id (no auth theo selection): domain token/id, compensating state (status visibility), SEC-02 + IDOR, schema.

Tổng AI: 40 TC → [API2 PoolB/generated.md](#).

4.2 Audit Summary

34 VALID / 2 INVALID / 4 INCOMPLETE → [audit.md](#).

Expected IDOR = **deny** (403/404), dù surface detail công khai theo selection.

4.3 Extended Test Cases

6 TC → [extended.md](#): oracle PII shipping_address, horizontal IDOR với token A đọc đơn B, admin trên detail user, dual-actor compare, 401 JSON, count sau checkout.

Excel: [API2 PoolB.xlsx](#).

4.4 Execution Results

Metric	Value
Requests	26
Assertions	35
Passed	30
Failed	5

- HTML: [newman-api2-fr11.html](#)
- CLI: [newman-api2-fr11-cli.txt](#)

4.5 Bugs Found

Bug	Severity	Issue	Evidence
BUG-ORDERS-001 IDOR + PII leak on GET /api/orders/:id	Critical	#302	BUG-ORDERS-001-idor.json, issue-302-*.png

5. API 3 — Pool C (FR-15)

5.1 AI Generation Process

Steps A–D cho POST/PUT/DELETE /api/products: domain (price > 0, name, ...), state (create→update→delete), SEC-02/SEC-03, schema.

Tổng AI: 40 TC → [API3 PoolC/generated.md](#).

5.2 Audit Summary

33 VALID / 3 INVALID / 4 INCOMPLETE → [audit.md](#).

Security expected = **401/403 theo spec**, không theo SUT hiện không có middleware.

5.3 Extended Test Cases

6 TC → [extended.md](#): unauth mutate success-as-bug, user vs admin, impact chain list sau create, quirk GET /api/products/:id sau DELETE.

Excel: [API3 PoolC.xlsx](#).

5.4 Execution Results

Metric	Value
Requests	22
Assertions	25
Passed	13
Failed	12

- HTML: [newman-api3-fr15.html](#)
- CLI: [newman-api3-fr15-cli.txt](#)

5.5 Bugs Found

Bug	Severity	Issue
BUG-ADMINPROD-001 CRUD thiếu JWT/admin	Critical	#303
BUG-ADMINPROD-002 Neg price / empty body accepted	Major	#304
BUG-ADMINPROD-003 DELETE missing id → 200	Minor	#305

Evidence: bug-reports/screenshots/issue-303|304|305-*.png + JSON/TXT dumps.

6. Postman Features Used

Feature	Cách dùng
Workspaces	Import collection + environment local
Collections	EShop-HW06.postman_collection.json (Setup, API1-3, Data-Driven, CI Smoke)
Variables / Environments	baseUrl, studentId, tokens, order/product ids
Pre-request scripts	Collection-level inject X-Student-Id: 23127152
Tests / assertions	Status, JSON schema, security oracles
Collection Runner + data file	products-search-data.csv → 8/8 pass
Examples / mock seed	postman/mock/fr05-list-example.json
Monitors	☐ — thay bằng GitHub Actions (Phase 5)

Phase 4 Newman:

Run	Result	Report
Data-driven FR-05	8 iter · 32 assert · 0 fail	newman-report-data-driven.html
Full collection	Failures = known Phase 1-3 bugs	newman-report.html

Chi tiết: [postman/PHASE4_NOTES.md](#).

7. CI/CD Integration

Workflow: [.github/workflows/hw06-api-tests.yml](#)

Báo cáo đầy đủ: [CI_CD_Report.md](#)

Sample	Commit	Actions run	Screenshot
All pass	8b44b66	32269053058	cicd/screenshots/run-all-pass.png
One fail	2fe51ee	32269286455	cicd/screenshots/run-one-fail.png

CI Smoke (Must Pass) tách khỏi suite security (expected fail = bug evidence). Intentional fail chỉ đổi ciExpectStatus 200→201.

8. Agent Skill — AI Test Generator (G9.5)

Thiết kế generator: **API spec** → **TC (domain/state/security/schema)** → **human audit** → **extend ≥5** → **Postman/Newman**.

Artifact	Path
Architecture diagram (self-drawn Mermaid → PNG)	agent-skill/diagram.png
Mermaid source	agent-skill/diagram.mmd
Pseudocode + design	agent-skill/pseudocode.md
Working pipeline skills	.agents/skills/README-HW06.md

Pipeline skills thực tế: api-test-generate → api-test-audit → api-test-extend → api-test-execute → api-test-cicd.

9. Conclusion

Hạng mục	Số liệu
TC AI generated	120 (40×3)
TC manual extended	18 (6×3)
Bugs filed (GitHub)	6 (#300–#305, Anhnguyenk835)
Newman reports	API1/2/3 + full + data-driven + CI artifact
CI samples	1 green + 1 intentional red

Bài học: AI sinh nhanh domain/schema nhưng thiếu oracle cụ thể (row-count, PII field, Content-Type trên lỗi) và dễ nhầm “SUT đang trả 200” với expected. Human audit + extend security là bước bắt buộc trước khi tin Newman fail là bug thật.

Appendix

Document	Path
AI Audit Report	ai-audit/AI Audit Report.md
AI Critique	ai-audit/AI Critique.md
Bug index	bug-reports/github issues links.md
Git commit log	git-commit-log.txt
Checklist	CHECKLIST.md